Persistence and environmental fate of synplastomic DNA in soil microcosms

1. Introduction

The fate of extracellular DNA in the environment is determined by a complex interplay of biotic and abiotic factors that can lead to its degradation or protection. Physical agents like heat and ultraviolet light, chemical factors such as pH, reactive oxygen species, heavy metals, as well as enzymatic hydrolysis by nucleases from plants and microorganisms, contribute to DNA alteration and degradation (Pontiroli et al., 2007). However, eDNA can be protected against this enzymatic degradation through its adsorption onto soil components. Soil is a chemically complex and spatially heterogeneous environment containing surface-reactive particles like clay, sand, silt, and humic substances that can adsorb nucleic acids. However, adsorption is considered partial and reversible, meaning changes in environmental conditions could lead to desorption, making the DNA accessible to bacteria and DNases. Previous studies have shown that DNA can persist in soil for extended periods, particularly when bound to soil colloids or protected within organic matrices that limit access to enzymatic degradation (Demaneche et al., 2001; Freeman et al., 2023; Romanowski et al., 1992). These protective interactions can extend the environmental persistence of eDNA and, consequently, enhance the likelihood of its uptake by naturally competent microorganisms.

The increasing use of genetically modified crops has raised critical questions about the environmental fate of transgenic DNA, particularly regarding its persistence and potential for horizontal gene transfer (HGT) in soil ecosystems. Natural transformation allows competent bacterial cells to bind and take up exogenous DNA from their surroundings, in some cases integrating homologous sequences into their genome or incorporating plasmid DNA (Nielsen & Townsend, 2004). While laboratory experiments have demonstrated that eDNA adsorbed by soil components can transform competent bacterial cells, the occurrence of natural transformation in soil in situ is generally thought to occur at very low frequencies (Nielsen et al., 1998). The persistence of transgenic plant DNA in soil can range from a few days to several years, influenced by environmental conditions that affect microbial and enzymatic activity (Gebhard & Smalla, 1999; Lee et al., 2010; Lerat et al., 2007). Although degradation typically occurs rapidly, a fraction of DNA can escape degradation and persist. For instance, a study using transplastomic tobacco DNA in soil microcosms demonstrated that although there was a dramatic initial decrease in the number of transgene sequences within the first two weeks, a small fraction escaped rapid degradation and was still detected after 4 years in soil samples amended with transplastomic leaves or purified DNA. Furthermore, DNA extracted from these soil microcosms after 4 years remained biologically active and could still transform competent bacterial cells in vitro (Pontiroli et al., 2010).

Synplastomic DNA, engineered for plastid-specific expression, exhibits several unique characteristics, such as its circular conformation and high copy number within plant chloroplasts, that may significantly impact its persistence and mobility within the soil environment (Occhialini et al., 2022). The circular conformation of synplastomic DNA confers inherent advantages, including enhanced stability and resistance to exonuclease degradation, as well as increased protection against environmental stressors, which may prolong its residence time in soil. While the degradation kinetics of recombinant DNA fragments from nuclear and plastid transgenic plants in soil have been documented, the behavior of episomal synplastomic DNA remains largely uncharacterized. This knowledge gap is significant, as synplastomic constructs may exhibit distinct interactions with soil matrices and microbial communities, influencing their environmental persistence and bioavailability for bacterial transformation. The present study addresses these knowledge gaps by evaluating the long-term persistence of synplastomic plant DNA in soil microcosms under controlled degradation conditions, simulating natural decay of plant residues and assessing the influence of input source and concentration by comparing degradation dynamics among intact leaves, ground leaves, and purified plant synplastomic DNA, each representing different levels of physical complexity and accessibility. This comparison will help determine whether structural features of plant inputs or loading rates influence DNA stability in soil. Through this work, we aim to provide a comprehensive understanding of synplastomic DNA persistence and degradation dynamics in soil environments. These insights are essential for informing biosafety assessments and guiding regulatory frameworks related to plastid-based genetic engineering technologies.

2. Methodology

Experimental Design and Data Simulation

A fully crossed factorial soil microcosm framework was simulated to evaluate the effects of DNA source type and initial DNA concentration on environmental DNA (eDNA) persistence dynamics in soil. Three DNA source types were considered: purified plasmid DNA, mechanically disrupted ground tissue, and intact leaf tissue, representing increasing levels of physical protection of extracellular DNA. Each source type was evaluated at three initial concentrations (500, 1000, and 2000 ng g^{-1} soil dry weight), generating nine source × concentration treatment combinations. Persistence dynamics were evaluated at seven sampling times (0, 1, 7, 14, 28, 90, and 180 days) using a destructive sampling design with three independent replicates per treatment-time combination (n = 189 total observations). Each microcosm was sampled once and discarded, ensuring independence among observations. Simulated copy number data were parameterized from the calibration framework of Sirois and Buckley (2019). Initial copy numbers were scaled proportionally to the higher DNA concentrations used in the present study, yielding approximately 5×10^{10} to 2×10^{11} gene copies g^{-1} soil at Day 0. A qPCR detection threshold of 2×10^{5} copies g^{-1} soil was applied. Replicate variability was simulated using lognormal error (coefficient of variation = 10%). Values below the detection limit were imputed at the threshold value. The response variable for all analyses was fraction remaining, calculated as the observed copy number at time t divided by the mean Day 0 copy number for the corresponding treatment group. Fraction remaining was $\log_{10}$-transformed prior to analysis.

Statistical Analysis

The effects of DNA source, initial concentration, and their interaction on fraction remaining were evaluated using two-way analysis of variance (ANOVA) at Day 7 and Day 180, representing the early decay phase and late-stage persistence phase, respectively. When significant interaction effects were detected, follow-up one-way ANOVAs were conducted separately within each DNA source. Pairwise comparisons were evaluated using Tukey's Honest Significant Difference (HSD) test with family-wise adjustment at $\alpha = 0.05$. Model assumptions were assessed using residual and quantile–quantile diagnostics.

Nonlinear Decay Modeling

eDNA persistence dynamics were modeled using nonlinear least squares regression based on the exponential decay model of Sirois and Buckley (2019):

$$y(t) = s + (1 - s)e^{-t\,e^{d}}$$

where $y(t)$is the fraction remaining at time t, $d$is the logarithm of the decay-rate multiplier, and $s$is the long-term stabilization asymptote. Larger values of e^{d}correspond to steeper decay trajectories, whereas $s$represents the residual fraction remaining after long-term stabilization.

Three competing model structures were evaluated: (i) a null model constraining both $d$and $s$to single global values, (ii) Model B allowing group-specific $s$values while constraining $d$globally, and (iii) Model A allowing both $d$and $s$to vary among all source × concentration groups. All models were fit simultaneously to the complete dataset (n = 189) using the nls() function in R to ensure comparability of likelihood-based model selection statistics.

Model Selection and Parameter Estimation

Model selection was performed using Akaike's Information Criterion (AIC) and Bayesian Information Criterion (BIC), with lower values indicating improved fit relative to model complexity where k is the number of estimated parameters and $n$is the number of observations. Parameter estimates and 95% confidence intervals were obtained from the selected global model using profile likelihood estimation (confint() in R). Negative stabilization asymptotes were interpreted as biologically implausible because stabilization fractions are constrained to non-negative values. All analyses were conducted in R.

3. Assumptions check

Normality of ANOVA residuals was evaluated using the Shapiro-Wilk test. Residuals were approximately normally distributed for all source × time point combinations (W range: 0.898–0.972, all $p > 0.05$), with the exception of leaf-derived DNA at Day 7 ($W = 0.831$, $p = 0.046$). Given the small sample size per group ($n = 9$), the limited power of the Shapiro-Wilk test, and the known robustness of the F-test to mild departures from normality under balanced designs, this borderline result was not considered to invalidate the ANOVA inference for this group.

Source	**Time point**	**W statistic**	**p-value**
Plasmid	Day 7	0.949	0.674
Plasmid	Day 180	0.898	0.241
Ground tissue	Day 7	0.914	0.344
Ground tissue	Day 180	0.972	0.915
Leaves	Day 7	0.831	**0.046**
Leaves	Day 180	0.893	0.213

Table 1. Shapiro-Wilk normality test results for ANOVA residuals within each DNA source at Day 7 and Day 180.

Visual inspection of diagnostic plots supported the Shapiro-Wilk results. At Day 7, residuals showed mild deviation from normality in the tails of the Q-Q plot, consistent with the borderline Shapiro-Wilk result for leaf-derived DNA ($W = 0.831$, $p = 0.046$), and a slight upward trend in the Scale-Location plot suggested mild heteroscedasticity at higher fitted values. At Day 180, diagnostic plots indicated that assumptions of normality and homogeneity of variance were adequately met, with residuals distributed randomly around zero and points closely following the Q-Q reference line. Given the balanced design and the robustness of the F-test to mild assumption violations, these departures were not considered sufficient to invalidate ANOVA inference at either time point.

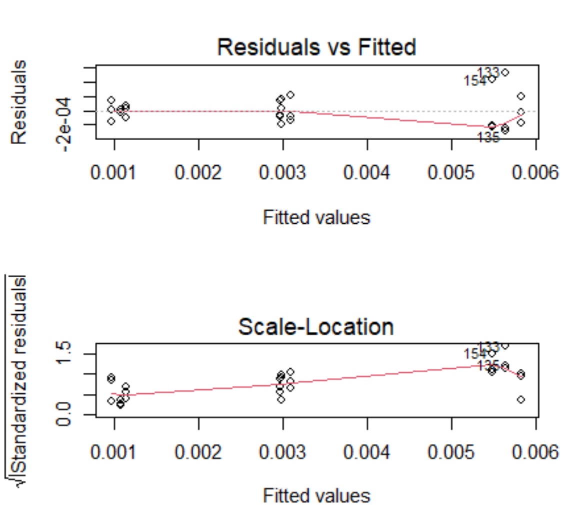

Fig 1. ANOVA diagnostic plots for eDNA fraction remaining at Day 7

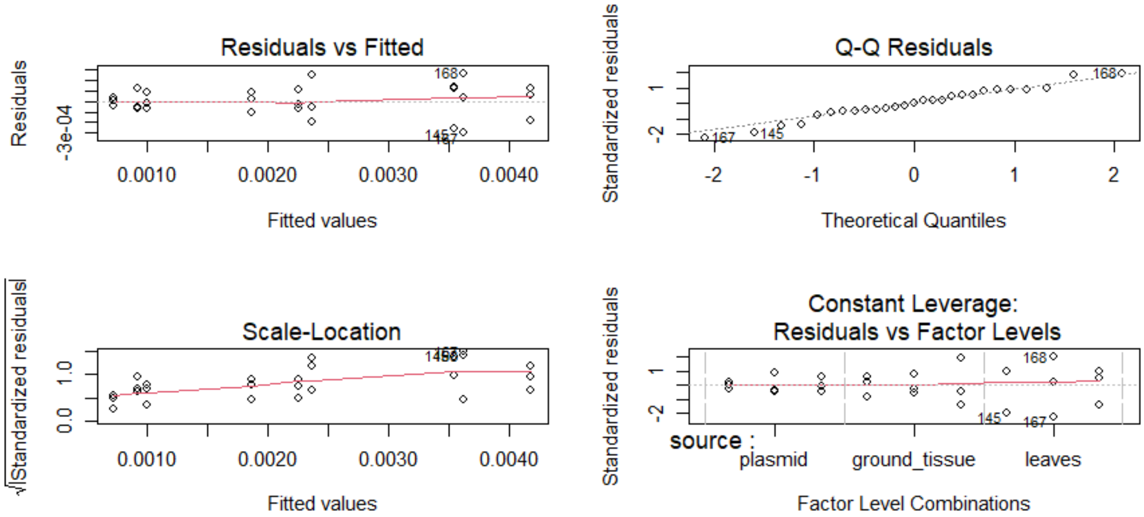

Fig 2. ANOVA diagnostic plots for eDNA fraction remaining at Day 180

Residual diagnostics for the selected global Model A indicated departures from normality (Shapiro–Wilk: $W = 0.629$, $p < 0.001$) and heteroscedasticity, with greater residual variance observed during the early phase of decay (Day 0–1) relative to later time points. These patterns are consistent with the large dynamic range characteristic of exponential decay processes spanning multiple orders of magnitude. Despite these deviations, residuals remained centered near zero across all source × concentration groups with no clear evidence of systematic group-specific bias. Because the primary objective of the nonlinear regression analysis was estimation of relative decay dynamics and comparison of competing model structures, the model was retained for inference. Likelihood-based model comparisons (AIC/BIC) are generally robust to moderate deviations from residual normality, particularly under balanced designs and large dynamic-range biological data.

4. Results

4.1. Source and concentration jointly affect eDNA decay dynamics

At Day 7, two-way ANOVA revealed significant main effects of DNA source ($F_{2,18} = 484.5$, $p < 0.001$), initial concentration ($F_{2,18} = 7.3$, $p = 0.005$), and a significant source × concentration interaction ($F_{4,18} = 7.8$, $p < 0.001$). At Day 180, all three effects remained significant: source ($F_{2,18} = 399.0$, $p < 0.001$), concentration ($F_{2,18} = 34.3$, $p < 0.001$), and their interaction ($F_{4,18} = 13.0$, $p < 0.001$) (Table 1). The significant interaction at both time points indicate that the effect of initial DNA concentration on persistence dynamics differed across DNA sources, justifying follow-up analyses within each DNA source independently.

Post-hoc comparisons (Tukey HSD, $\alpha = 0.05$) at Day 7 revealed that all three DNA sources differed significantly from one another in fraction remaining (all $p < 0.001$), with the ordering leaves > ground tissue > plasmid. Leaf-derived DNA retained the greatest fraction remaining, whereas purified plasmid DNA retained the least. Concentration effects at Day 7 were less pronounced, only the comparison between 2000 and 500 ng g^{-1} reached statistical significance ($\Delta = 6.22 \times 10^{-4}$, $p = 0.004$), while comparisons between 1000 and 500 ng g^{-1} ($p = 0.411$) and between 2000 and 1000 ng g^{-1} ($p = 0.060$) were not significant.

		Day 7		Day 180	
Source of variation	df	F	p	F	p
DNA source	2	484.51	< 0.001	398.97	< 0.001
Concentration	2	7.30	0.005	34.26	< 0.001
Source × Concentration	4	7.84	< 0.001	12.99	< 0.001

Table 1. Effects of DNA source, initial concentration, and their interaction on eDNA fraction remaining at Day 7 and Day 180.

4.2. Effects of concentration within each DNA source (follow-up ANOVA per source)

Follow-up one-way ANOVAs conducted separately within each DNA source showed contrasting concentration responses between Day 7 and Day 180 (Table 2). At Day 7, no significant concentration effects were detected for plasmid DNA ($F_{2,6} = 1.27$, $p = 0.348$). Similarly, no significant concentration differences were detected for ground tissue DNA at Day 7 ($F_{2,6} = 0.41$, $p = 0.680$), suggesting no detectable concentration effect on early decay dynamics under the present experimental design. In contrast, leaf-derived DNA showed a significant concentration effect at Day 7 ($F_{2,6} = 11.16$, $p = 0.010$).

Pairwise comparisons within sources (Tukey HSD; Table 3) showed that plasmid DNA exhibited no significant differences among concentration levels at Day 7 (all $p > 0.566$). Similarly, no significant pairwise differences were detected for ground tissue DNA at Day 7 (all $p > 0.658$). In contrast, leaf-derived DNA showed significant differences among concentrations, with the 2000 ng g^{-1} treatment retaining significantly greater fractions than both 500 ng g^{-1} ($\Delta = 1.83 \times 10^{-3}$, $p = 0.010$) and 1000 ng g^{-1} ($\Delta = 1.42 \times 10^{-3}$, $p = 0.030$). These results indicate that concentration-dependent differences were detectable in leaf-derived DNA during the early phase of decay.

These patterns were largely consistent at Day 180, although concentration effects became more pronounced. One-way ANOVA showed significant effects of concentration for plasmid ($F_{2,6}$ = 11.43, p = 0.009), ground tissue ($F_{2,6}$ = 13.35, p = 0.006), and leaf-derived DNA ($F_{2,6}$ = 21.00, p = 0.002). All pairwise source comparisons remained highly significant (all p < 0.001), maintaining the ordering leaves > ground tissue > plasmid in the fraction remaining. In the crossed ANOVA, all pairwise concentration comparisons were also significant at Day 180 (2000 vs. 500 ng g^{-1}: p < 0.001; 1000 vs. 500 ng g^{-1}: p < 0.001; 2000 vs. 1000 ng g^{-1}: p = 0.025), indicating an overall concentration effect during the long-term phase of persistence. However, the significant source × concentration interaction indicates that the magnitude of this effect differed among DNA sources. Follow-up within-source analyses showed that plasmid DNA exhibited no detectable concentration differences at Day 180 (all p ≥ 0.904), whereas concentration effects were more evident in leaf-derived DNA, which consistently retained the greatest fraction remaining across all treatments.

	Day 7		**Day 180**	
DNA source	F	p	F	p
Plasmid	1.27	0.348	11.43	0.009
Ground tissue	0.41	0.680	13.35	0.006
Leaves	11.16	0.010	21.00	0.002

Table 2. Effects of initial DNA concentration on fraction remaining within each DNA source at Day 7 and Day 180 (one-way ANOVA, df = 2, 6 for all tests).

DNA source	**Comparison**	**Day 7 Δ**	**p value**	**Day 180 Δ**	**p value**
Plasmid	1000 vs. 500 ng g^{-1}	9.40×10^{-5}	0.566	3.31×10^{-4}	0.010 *
Plasmid	2000 vs. 500 ng g^{-1}	1.37×10^{-4}	0.333	2.69×10^{-4}	0.025 *
Plasmid	2000 vs. 1000 ng g^{-1}	4.31×10^{-5}	0.879	-6.29×10^{-5}	0.686
Ground tissue	1000 vs. 500 ng g^{-1}	1.43×10^{-4}	0.860	2.43×10^{-4}	0.072
Ground tissue	2000 vs. 500 ng g^{-1}	-1.01×10^{-4}	0.927	4.52×10^{-4}	0.005 **
Ground tissue	2000 vs. 1000 ng g^{-1}	-2.44×10^{-4}	0.658	2.09×10^{-4}	0.118
Leaves	1000 vs. 500 ng g^{-1}	4.10×10^{-4}	0.599	1.25×10^{-3}	0.021 *
Leaves	2000 vs. 500 ng g^{-1}	1.83×10^{-3}	0.010 **	2.10×10^{-3}	0.002 **
Leaves	2000 vs. 1000 ng g^{-1}	1.42×10^{-3}	0.030 *	8.58×10^{-4}	0.087

Significance codes: *** p < 0.001, ** p < 0.01, * p < 0.05

Table 3. Pairwise comparisons (Tukey HSD) for the effect of initial DNA concentration within each DNA source on fraction remaining at Day 7 and Day 180.

4.3. Mechanistic decay parameters

The null model estimated a significant global decay parameter ($d = 0.050 \pm 0.019$, $p = 0.010$; $\exp(d) = 1.051$), indicating detectable eDNA degradation across all treatment groups, but a non-significant stabilization asymptote ($s = 0.0025 \pm 0.003$, $p = 0.416$). This suggests that pooling all source × concentration groups into a single shared decay trajectory reduces biologically relevant variation in long-term persistence dynamics. Model B, which fixed d at a single shared value while allowing s to vary among groups, provided little improvement in fit over the null model (residual standard error, RSE = 0.0351 for both models) and produced several negative s estimates. Because negative stabilization fractions are biologically implausible, these results indicate that constraining all treatment groups to a single shared decay parameter produces compensatory parameter behavior and inadequately represents the decay process in these data. Model B was therefore rejected on both statistical and biological grounds.

DNA source	Concentration (ng g^{-1})	d	SE (d)	p (d)	exp(d)	s	SE (s)	p (s)
Plasmid	500	0.164	0.046	0.046	1.177	7.64×10^{-4}	7.26×10^{-3}	0.916
	1000	0.256	0.047	< 0.001	1.292	9.64×10^{-4}	7.25×10^{-3}	0.884
	2000	0.306	0.048	< 0.001	1.359	1.08×10^{-3}	7.25×10^{-3}	0.875
Ground tissue	500	−0.153	0.046	0.026	0.858	1.73×10^{-3}	7.28×10^{-3}	0.826
	1000	0.041	0.047	0.002	1.041	2.33×10^{-3}	7.25×10^{-3}	0.770
	2000	0.133	0.047	< 0.001	1.143	2.55×10^{-3}	7.25×10^{-3}	0.754
Leaves	500	−0.199	0.046	0.005	0.819	3.26×10^{-3}	7.28×10^{-3}	0.672
	1000	−0.190	0.046	0.002	0.827	3.49×10^{-3}	7.28×10^{-3}	0.566
	2000	−0.061	0.046	0.020	0.940	4.51×10^{-3}	7.28×10^{-3}	0.473

d = natural logarithm of the decay rate, exp(d) = decay rate multiplier (values > 1 indicate faster-than-reference decay; values < 1 indicate slower-than-reference decay). s = long-term stabilization asymptote. SE = standard error. p values derived from global Model A joint estimation.

Table 4. Group-specific parameter estimates from the selected decay model (Model A) for each DNA source × initial concentration combination, estimated jointly from all 189 observations using nonlinear least squares regression.

Model A substantially improved fit relative to both alternative structures (RSE = 0.0281), representing approximately a 20% reduction in residual error relative to the null model. The model retained 171 residual degrees of freedom after estimation of 18 group-specific parameters. Across all treatment combinations, fitted trajectories showed a rapid initial decline in eDNA abundance followed by stabilization at a low but detectable plateau throughout the remainder of the 180-day experiment (Figs. 1–3). Fraction remaining declined from approximately 1.0 at Day 0 to values near 10^{-3} by Day 7 in all three DNA sources, indicating that the majority of degradation occurred during the earliest phase of the experiment. After Day 7, fitted curves approached near-asymptotic behavior with comparatively limited additional decline through Day 180 (Figs. 1–3). Consistent with these trajectories, all nine group-specific decay parameter estimates were statistically significant ($p < 0.05$), whereas none of the nine stabilization asymptote estimates differed significantly from zero individually (all $p > 0.47$), reflecting the difficulty of precisely estimating very small asymptotic fractions with only three replicates per treatment group.

The decay rate multiplier exp(d) showed clear source-dependent structuring across treatment groups. Purified plasmid DNA exhibited exp(d) values consistently greater than unity across all concentration levels (1.098–1.320), with values increasing with initial DNA concentration. This pattern indicates progressively steeper early decay trajectories with increasing DNA concentration in plasmid-derived eDNA. In contrast, leaf-derived DNA showed exp(d) values consistently below unity across all concentration levels (range: 0.865–0.897), with comparatively little variation among concentrations, suggesting reduced sensitivity of early decay dynamics to initial DNA load in intact leaf tissue. Ground tissue DNA exhibited intermediate behavior, with exp(d) shifting from below unity at 500 ng g^{-1} (exp(d) = 0.902) to above unity at 1000 and 2000 ng g^{-1} (exp(d) = 1.158 and 1.237, respectively). Correspondingly, fitted trajectories for ground tissue DNA stabilized at intermediate levels between plasmid and intact leaf tissue (Fig. 2), consistent with partial physical protection by disrupted cellular material at lower DNA loads.

Although individual stabilization asymptote estimates (s) did not reach statistical significance, a consistent biological trend was observed across all concentration levels: s increased from purified plasmid DNA (7.6×10^{-4} to 1.1×10^{-3}) through mechanically disrupted ground tissue (1.6×10^{-3} to 2.3×10^{-3}) to intact leaf-derived DNA (3.1×10^{-3} to 5.2×10^{-3}), leaf-derived DNA exhibiting the largest asymptotes, followed by ground tissue and plasmid DNA. Within each source, s also increased with initial DNA concentration, consistent with the significant concentration effects detected at Day 180 by one-way ANOVA (plasmid: $F_{2,6} = 11.43$, $p = 0.009$; ground tissue: $F_{2,6} = 13.35$, $p = 0.006$; leaves: $F_{2,6} = 21.00$, $p = 0.002$). Across all treatment combinations, estimated stabilization fractions remained below 0.006, indicating that the majority of added eDNA degraded during the experimental period.

Model selection was performed using the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC), both of which strongly favored Model A (AIC = −795.2, BIC = −733.6) over Model B (AIC = −718.0, BIC = −682.4) and the null model (AIC = −725.7, BIC = −715.9). Relative to the null model, Model A reduced AIC by 69.5 units, indicating substantially improved fit despite the larger number of estimated parameters (Table 3).

Model	**Parameters (k)**	**RSE**	**df**	**AIC**	**ΔAIC**	**BIC**	**ΔBIC**
Null	2	0.0351	187	−725.7	69.5	−715.9	17.7
Model B	10	0.0351	179	−718.0	77.2	−682.4	51.2
Model A	18	0.0281	171	−795.2	—	−733.6	—

Table 5. Model selection criteria for three competing exponential decay models fit to eDNA fraction remaining data (n = 189 observations across nine source × concentration treatment groups).

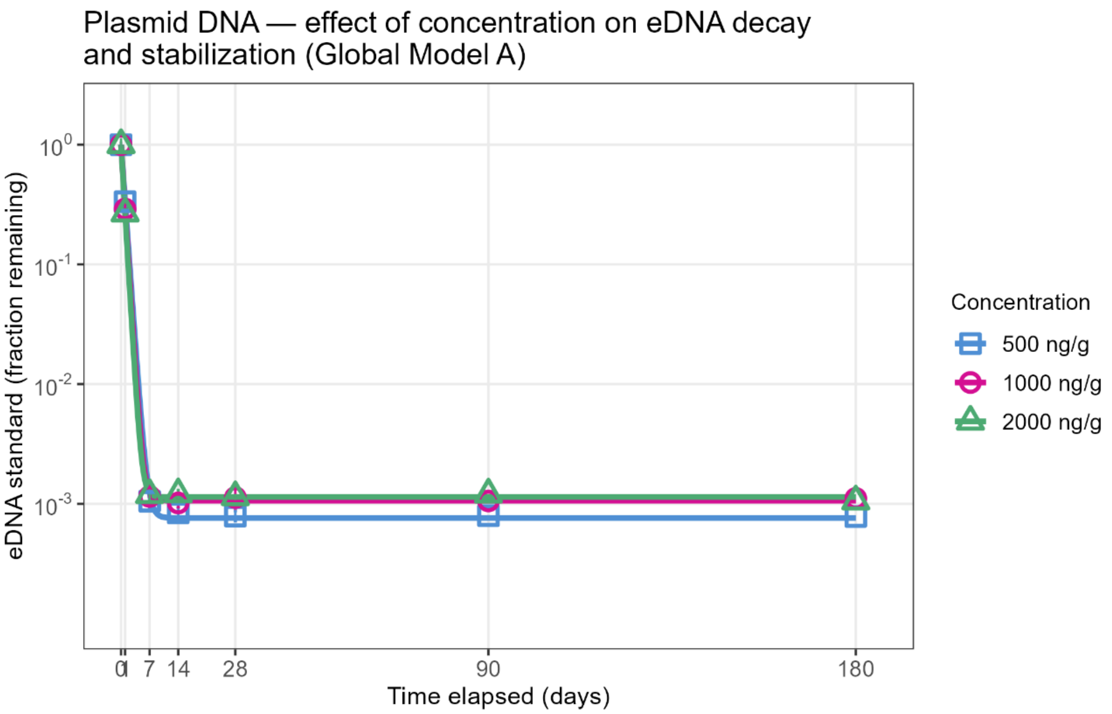

Fig 1. Decay trajectories of plasmid-derived synplastomic DNA across three initial DNA concentrations (500, 1000, and 2000 ng g^{-1} soil) fitted using Global Model A.

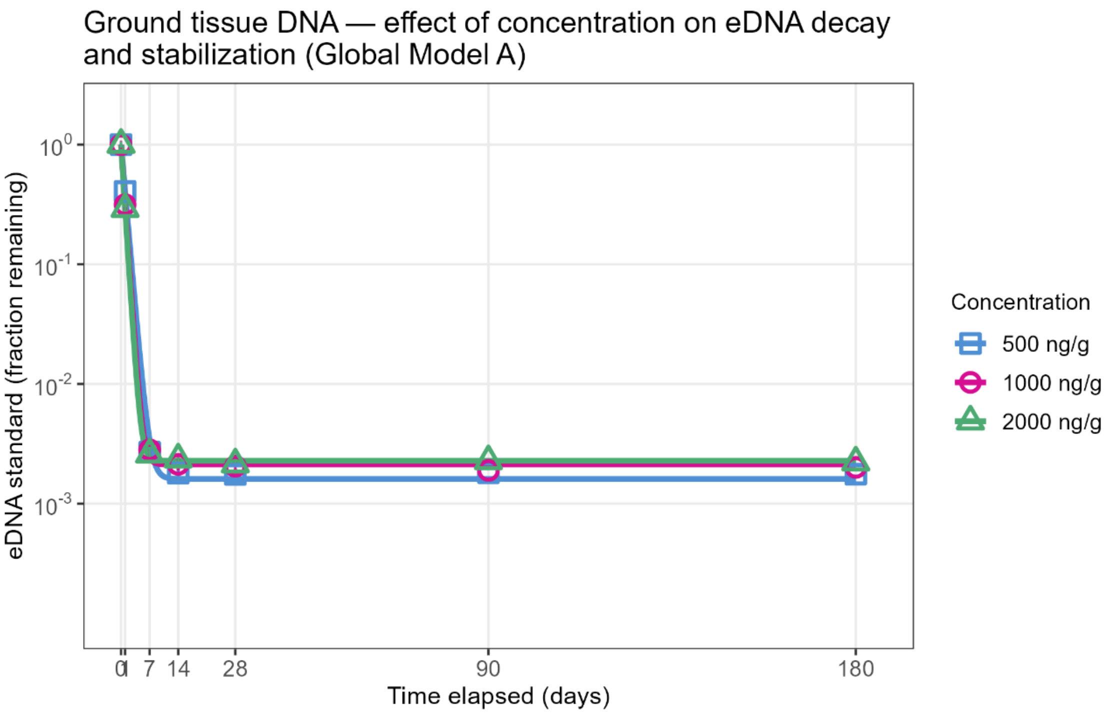

Fig. 2. Decay trajectories of ground tissue-derived synplastomic DNA across three initial DNA concentrations (500, 1000, and 2000 ng g^{-1} soil) fitted using Global Model A.

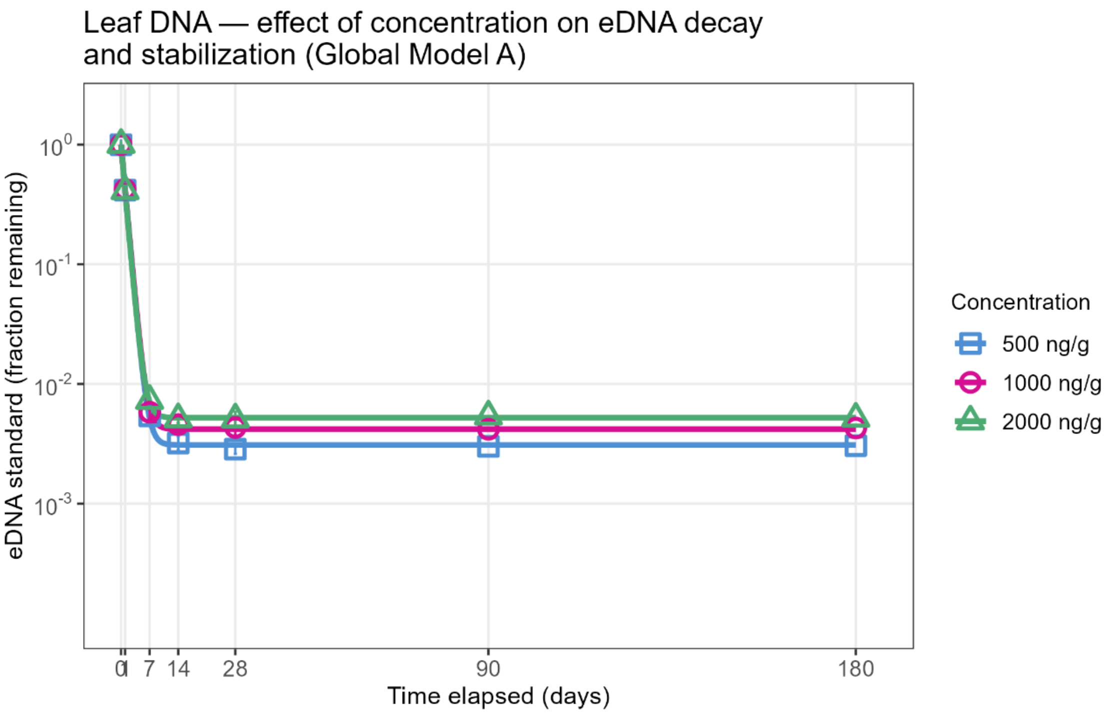

Fig. 3. Decay trajectories of ground tissue-derived synplastomic DNA across three initial DNA concentrations (500, 1000, and 2000 ng g^{-1} soil) fitted using Global Model A.

5. Discussion

The present study demonstrates that both DNA source and initial DNA concentration influence eDNA persistence dynamics in soil, but that these effects are strongly source dependent. Across all analyses, intact leaf-derived DNA consistently exhibited the greatest persistence, mechanically disrupted ground tissue showed intermediate persistence, and purified plasmid DNA degraded most rapidly. These patterns were evident both in the ANOVA-based comparisons at discrete sampling points and in the mechanistic decay modeling, indicating that the physical state of DNA strongly modulates degradation dynamics in soil microcosms.

The significant source × concentration interaction detected at both Day 7 and Day 180 indicates that the effect of initial DNA concentration cannot be interpreted independently of DNA source. Concentration effects were weak or absent in purified plasmid DNA during the early phase of decay, whereas intact leaf-derived DNA showed detectable concentration-dependent differences even at Day 7. This suggests that increasing DNA load alone does not universally enhance persistence, but rather that the influence of concentration depends on the structural context in which DNA is introduced into soil.

The consistently greater persistence of leaf-derived DNA likely reflects protection provided by intact plant cellular structures. DNA embedded within intact tissue may be partially shielded from extracellular nucleases, microbial degradation, and adsorption–desorption processes immediately after introduction into soil. In contrast, purified plasmid DNA lacks physical protection and is directly exposed to enzymatic degradation and sorption dynamics, which may explain the steeper initial decay trajectories observed in the plasmid treatments. Ground tissue DNA exhibited intermediate behavior, consistent with partial disruption of protective cellular structures during homogenization. Together, these findings support the hypothesis that tissue integrity contributes substantially to eDNA stabilization in soil environments.

All treatment groups exhibited rapid initial declines in detectable eDNA abundance, with fraction remaining decreasing from approximately 1.0 at Day 0 to values near 10^{-3} by Day 7. After this early decline, decay trajectories approached near-asymptotic behavior and remained relatively stable through Day 180. This biphasic pattern is consistent with previous studies reporting an initial phase of rapid degradation followed by persistence of a small stabilized DNA fraction associated with soil particles or physically protected microenvironments. The fitted stabilization asymptotes from Model A were uniformly small (< 0.006), indicating that more than 99% of the added eDNA degraded during the experiment regardless of source or concentration. Nevertheless, the ordering of asymptotic values (leaves > ground tissue > plasmid) suggests that DNA source influences not only early degradation rates but also the magnitude of the long-term stabilized fraction.

Mechanistic modeling further clarified differences among DNA sources. The decay rate multiplier exp(d) revealed systematic source-dependent structure across treatment groups. Purified plasmid DNA exhibited exp(d) values consistently greater than unity, with values increasing alongside initial concentration, indicating progressively steeper early decay trajectories at higher DNA loads. In contrast, leaf-derived DNA showed exp(d) values consistently below unity and comparatively stable across concentrations, suggesting slower relative decay dynamics and reduced sensitivity to concentration. Ground tissue DNA again exhibited intermediate behavior, transitioning from below-unity exp(d) values at low concentration to above-unity values at higher concentrations. These trends suggest that the mechanisms governing synplastomic DNA degradation differ among physical DNA states and may shift with increasing DNA abundance.

Although concentration effects became statistically stronger at Day 180, their biological magnitude remained relatively small compared with differences among DNA sources. This distinction is important because statistically significant concentration effects do not necessarily imply large ecological differences in persistence. The strongest and most consistent determinant of persistence throughout the experiment was the structural origin of the DNA itself. Consequently, studies evaluating eDNA degradation in soil should carefully consider whether experimentally introduced DNA realistically represents naturally occurring biological material. Experiments relying exclusively on purified DNA may underestimate persistence relative to DNA associated with intact tissue or cellular material.

Several limitations should also be considered when interpreting these findings. First, stabilization asymptote estimates were associated with large standard errors, and none differed significantly from zero individually, reflecting limited precision in estimating very small residual fractions with only three replicates per treatment group. Second, the experiments were conducted under controlled microcosm conditions that do not fully capture the environmental heterogeneity of natural soils, including fluctuating moisture, temperature, microbial activity, and physical disturbance. Third, the decay model assumes a shared functional form across all treatment groups, which may simplify potentially more complex degradation dynamics occurring at intermediate time scales. Despite these limitations, the combined ANOVA and nonlinear modeling framework provided internally consistent evidence that synplastomic DNA persistence in soil is jointly shaped by DNA source and concentration, with source-dependent physical protection emerging as the dominant factor controlling degradation dynamics. These findings have implications for environmental DNA monitoring, horizontal gene transfer studies, and risk assessments involving transgenic plant material, because they suggest that the persistence of extracellular DNA in soil depends strongly on the biological context in which that DNA enters the environment.

6. Conclusions

This study demonstrates that eDNA persistence in soil is jointly influenced by the DNA source and the initial DNA concentration. However, the magnitude and direction of the concentration effects depend strongly on the physical form of DNA. Across all analyses, intact leaf-derived DNA consistently exhibited the greatest persistence, mechanically disrupted ground tissue showed intermediate persistence, and purified plasmid DNA degraded most rapidly. All treatment groups experienced rapid initial degradation during the first seven days, followed by stabilization of a small residual fraction that remained detectable throughout the 180-day experiment. The significant source × concentration interaction observed at both Day 7 and Day 180 indicates that concentration effects cannot be interpreted independently of DNA source. Concentration-associated differences were most evident in leaf-derived DNA, whereas purified plasmid DNA showed comparatively limited responses to increasing DNA load, particularly during the early phase of decay.

Mechanistic decay modeling further demonstrated that a fully parameterized model allowing both decay rate and stabilization asymptote to vary among source × concentration groups provided the best overall description of the data. The fitted trajectories and parameter estimates suggest that intact plant tissue provides partial protection against degradation, likely by reducing direct exposure of DNA to extracellular nucleases and microbial activity within soil. Although all estimated stabilization fractions were small, indicating that most added eDNA degraded during the experiment, the consistent ordering of leaves, ground tissue, and plasmid across all concentrations supports the conclusion that the structural origin of DNA is a dominant determinant of persistence dynamics in soil microcosms. These findings have implications for synplastomic DNA monitoring and studies of extracellular gene persistence because experiments relying exclusively on purified DNA may underestimate the persistence of DNA associated with intact biological material under environmentally relevant conditions.

References

Demaneche, S., Jocteur-Monrozier, L., Quiquampoix, H., & Simonet, P. (2001). Evaluation of biological and physical protection against nuclease degradation of clay-bound plasmid DNA. *Appl Environ Microbiol*, *67*(1), 293-299. https://doi.org/10.1128/AEM.67.1.293-299.2001

Freeman, C. L., Dieudonne, L., Agbaje, O. B. A., Zure, M., Sanz, J. Q., Collins, M., & Sand, K. K. (2023). Survival of environmental DNA in sediments: Mineralogic control on DNA taphonomy. *Environmental DNA*, *5*(6), 1691-1705. https://doi.org/10.1002/edn3.482

Gebhard, F., & Smalla, K. (1999). Monitoring field releases of genetically modified sugar beets for persistence of transgenic plant DNA and horizontal gene transfer. *Fems Microbiology Ecology*, *28*(3), 261-272. https://doi.org/10.1111/j.1574-6941.1999.tb00581.x

Lee, B., Park, J. Y., Park, K. W., Harn, C. H., Kim, H. M., & Kim, C. G. (2010). Evaluating the Persistence of DNA from Decomposing Transgenic Watermelon Tissues in the Field. *Journal of Plant Biology*, *53*(5), 338-343. https://doi.org/10.1007/s12374-010-9121-z

Lerat, S., Gulden, R. H., Hart, M. A., Powell, J. R., England, L. S., Pauls, K. P., Swanton, C. J., Klironomos, J. N., & Trevors, J. T. (2007). Quantification and persistence of recombinant DNA of roundup ready corn and soybean in rotation. *Journal of Agricultural and Food Chemistry*, *55*(25), 10226-10231. https://doi.org/10.1021/jf072457z

Nielsen, K. M., Bones, A. M., Smalla, K., & van Elsas, J. D. (1998). Horizontal gene transfer from transgenic plants to terrestrial bacteria - a rare event? *Fems Microbiology Reviews*, *22*(2), 79-103. https://doi.org/10.1016/s0168-6445(98)00009-6

Nielsen, K. M., & Townsend, J. P. (2004). Monitoring and modeling horizontal gene transfer. *Nature Biotechnology*, *22*(9), 1110-1114. https://doi.org/10.1038/nbt1006

Occhialini, A., Pfotenhauer, A. C., Li, L., Harbison, S. A., Lail, A. J., Burris, J. N., Piasecki, C., Piatek, A. A., Daniell, H., Stewart, C. N., & Lenaghan, S. C. (2022). Mini-synplastomes for plastid genetic engineering. *Plant Biotechnology Journal*, *20*(2), 360-373. https://doi.org/10.1111/pbi.13717

Pontiroli, A., Ceccherini, M. T., Pote, J., Wildi, W., Kay, E., Nannipieri, P., Vogel, T. M., Simonet, P., & Monier, J. M. (2010). Long-term persistence and bacterial transformation potential of transplastomic plant DNA in soil. *Research in Microbiology*, *161*(5), 326-334. https://doi.org/10.1016/j.resmic.2010.04.009

Pontiroli, A., Simonet, P., Frostegard, A., Vogel, T. M., & Monier, J. M. (2007). Fate of transgenic plant DNA in the environment. *Environ Biosafety Res*, *6*(1-2), 15-35. https://doi.org/10.1051/ebr:2007037

Romanowski, G., Lorenz, M. G., Sayler, G., & Wackernagel, W. (1992). PERSISTENCE OF FREE PLASMID DNA IN SOIL MONITORED BY VARIOUS METHODS, INCLUDING A TRANSFORMATION ASSAY. *Applied and Environmental Microbiology*, *58*(9), 3012-3019. <Go to ISI>://WOS:A1992JL63200045

Sirois, S. H., & Buckley, D. H. (2019). Factors governing extracellular DNA degradation dynamics in soil. *Environ Microbiol Rep*, *11*(2), 173-184. https://doi.org/10.1111/1758-2229.12725